NEOfixer consensus disagreements — root-cause analysis

Generated: 2026-05-04 from the NEO Consensus snapshot built off NEOfixer cache `2026-05-04T13:04:11Z`

Summary

The NEO Consensus tab of the Planetary Defense Dashboard reports per-object membership across six independent NEO sources. Two disagreement classes involve NEOfixer:

- **121 objects** present in all five other sources (MPC NEA.txt, `mpc_orbits`, JPL CNEOS, ESA NEOCC, Lowell `astorb`) but not flagged `in_neofixer`.
- **38 objects** present in NEOfixer but in none of the other five.

Both populations are dominated by NEO-boundary effects in NEOfixer's Find_Orb solutions, plus a smaller residual of objects that are genuinely outside NEOfixer's catalog. Findings in detail below.

Group	Count	Root cause
A. Missing — in NEOfixer's `site=500` list, $q > 1.3$ (filtered by our code)	45	Our client-side `q > 1.3` filter in `lib/neo_consensus_neofixer.py`
B1. Missing — in NEOfixer's `/orbit/` catalog but not in tonight's target list	15	NEOfixer service drops these from `site=500` (priority/visibility); orbits still exist
B2. Missing — genuinely unknown to NEOfixer	61	NEOfixer's API returns "Unknown object" on both `/orbit/` and `/obs/`
C. Only-in-NEOfixer — boundary q	38	NEOfixer's Find_Orb $q \le 1.3$, MPC's q just over 1.3

Methodology

Inputs:

- `css_neo_consensus.v_membership_wide` (Gizmo replica, refreshed 2026-05-04)
- The NEOfixer JSON cache that today's ingest consumed: `app/.neofixer_targets.json` from `https://neofixerapi.a` generated 2026-05-04T13:04:11Z, 42,412 objects.
- Live per-object probes against `https://neofixerapi.arizona.edu/orbit/?object={packed}` and `https://neofixerapi.arizona.edu/obs/?object={packed}` for the 76 designations not found in the bulk cache.
- `mpc_orbits.q,e,i` (joined on the unpacked primary provisional designation).

The ingestor (`lib/neo_consensus_neofixer.py`) applies two client-side filters to the NEOfixer bulk response: drop rows with `neocp=True` and drop rows with `q > 1.3`. Both filters are documented in the module docstring; this analysis confirms that **0 of the 121 disagreements are NEOCP-related** and **45 (37%) are q-related**.

Group A — 45 with NEOfixer q > 1.3 (in cache, filtered by our code)

These are present in the NEOfixer `site=500` bulk response (i.e. NEOfixer itself lists them as NEO observation targets) but NEOfixer's Find_Orb q exceeds 1.3 AU. Our code at `lib/neo_consensus_neofixer.py:99–102` drops them, even though `mpc_orbits` (and the other four sources) classify them as NEOs.

q-distribution:

NF q range	Count
$1.30 < q \le 1.31$ (boundary)	17

NF q range	Count
1.31 < q ≤ 1.40	15
1.40 < q ≤ 1.50	4
q > 1.50	9

Largest INF q – MPC ql divergences (different orbit fits, possibly different epochs):

packed	designation	NF q	MPC q	Δ
K04BG0P	2004 BP160	2.4410	1.1958	+1.2452
K16Ea2U	2016 EU362	1.3530	0.6649	+0.6881
K17FE6Q	2017 FQ146	1.7490	1.1850	+0.5640
K05UF0F	2005 UF150	1.5810	1.0752	+0.5058
K14J79R	2014 JR79	1.5390	1.0440	+0.4950
K10R16E	2010 RE16	1.7290	1.2571	+0.4719
K10VK1Z	2010 VZ201	1.4240	1.0192	+0.4048
K14Qh9P	2014 QP439	1.6200	1.2968	+0.3232
K14HJ5T	2014 HT195	1.4830	1.2294	+0.2536
K17DC0B	2017 DB120	1.3670	1.1157	+0.2513

Full Group A list (sorted by NF q):

packed	designation	NF q	NF H	MPC q	MPC e
K06D62V	2006 DV62	1.3010	20.14	1.2982	0.5145
K14F00V	2014 FV	1.3010	20.23	1.3000	0.5124
K19N050	2019 NO5	1.3010	21.82	1.2978	0.3966
K19R86C	2019 RC86	1.3010	20.36	1.2981	0.3982
K03G00J	2003 GJ	1.3020	19.75	1.2896	0.5160
K18A13B	2018 AB13	1.3020	22.18	1.2992	0.4838
K19N03F	2019 NF3	1.3020	20.02	1.2981	0.4021
K20O07K	2020 OK7	1.3020	21.81	1.2997	0.4378
K22J01Z	2022 JZ1	1.3020	23.11	1.2968	0.3927
K07M24K	2007 MK24	1.3030	20.21	1.2990	0.3870
K11S21Z	2011 SZ21	1.3030	17.52	1.2945	0.5604
K20K05T	2020 KT5	1.3030	19.31	1.2983	0.4088
K23O44Q	2023 OQ44	1.3030	21.13	1.2994	0.4224
K22K11H	2022 KH11	1.3040	19.84	1.2957	0.4454
K03G29V	2003 GV29	1.3050	19.45	1.2952	0.5438
K03U26W	2003 UW26	1.3050	19.18	1.2952	0.5224
K16GM0O	2016 GO220	1.3100	17.50	1.2918	0.5881
K13HF0U	2013 HU150	1.3110	21.46	1.2721	0.4969
K14Qa3X	2014 QX363	1.3110	19.17	1.2886	0.5062
K17W11W	2017 WW11	1.3120	18.72	1.2947	0.5065
K23U09P	2023 UP9	1.3150	17.73	1.2986	0.5591
K11H52Y	2011 HY52	1.3160	20.18	1.2865	0.5390
K13T69G	2013 TG69	1.3160	20.77	1.2904	0.6135
K14Hq3K	2014 HK523	1.3160	22.46	1.2546	0.4095
K21H12M	2021 HM12	1.3180	21.17	1.2990	0.5183
K14Od2Q	2014 OQ392	1.3290	22.24	1.2949	0.4702
K15KF7Z	2015 KZ157	1.3300	23.49	1.2250	0.0618
K11K09K	2011 KK9	1.3410	22.32	1.2436	0.3909
K15MD1A	2015 MA131	1.3500	21.85	1.2868	0.4548
K16Ea2U	2016 EU362	1.3530	19.08	0.6649	0.2233
K15RI3S	2015 RS183	1.3670	16.34	1.2502	0.5815
K17DC0B	2017 DB120	1.3670	22.46	1.1157	0.4683
K12B86K	2012 BK86	1.4200	21.84	1.2711	0.3079

packed	designation	NF q	NF H	MPC q	MPC e
K10VK1Z	2010 VZ201	1.4240	20.81	1.0192	0.1904
K15HI2C	2015 HC182	1.4440	23.12	1.2902	0.0893
K14HJ5T	2014 HT195	1.4830	23.48	1.2294	0.4804
K14J79R	2014 JR79	1.5390	22.62	1.0440	0.2068
K05UF0F	2005 UF150	1.5810	20.28	1.0752	0.5109
K14Qh9P	2014 QP439	1.6200	20.15	1.2968	0.4072
K10R16E	2010 RE16	1.7290	19.17	1.2571	0.1488
K17FE6Q	2017 FQ146	1.7490	18.88	1.1850	0.3629
K25Oo9Z	2025 OZ509	2.0540	20.64	2.0516	0.1652
K25Oh4X	2025 OX434	2.2520	19.52	2.2623	0.0236
K04BG0P	2004 BP160	2.4410	19.28	1.1958	0.6412
K25Oq9X	2025 OX529	3.0420	17.90	2.9344	0.0715

Group B1 — 15 in NEOfixer's orbit catalog but not in today's `site=500` list

NEOfixer responds with valid orbit data for these designations on the per-object `/orbit/?object=` endpoint, but they do not appear in the bulk `site=500` target list (neither in today's cached snapshot nor in a fresh fetch without `q-max`). The most plausible interpretation is that NEOfixer's nightly priority generator filters them out — low priority, lost, behind the sun, or already heavily observed. Their orbits exist; they are simply not on tonight's observation list.

packed	designation	MPC q	NF /orbit/ q
K03YD6H	2003 YH136	0.2534	0.2531
K05X04Y	2005 XY4	0.4247	0.4235
K09V09P	2009 VP9	1.2718	1.2730
K10N81T	2010 NT81	0.9605	0.9605
K10V72D	2010 VD72	0.7519	0.7518
K10V21P	2010 VP21	0.3343	0.3343
K12F10D	2012 FD10	1.2429	1.2790
K15G00Y	2015 GY	0.8543	0.8541
K17Q17P	2017 QP17	0.6224	0.6225
K24R12P	2024 RP12	1.0011	1.0011
K25D49K	2025 DK49	1.0482	1.0479
K25O13O	2025 OO13	0.8695	0.8708
K25QE7N	2025 QN147	1.7482	1.7296
K25X06N	2025 XN6	0.5889	0.5952
K26B02E	2026 BE2	1.0921	1.0921

Note that several of these have very low MPC q values (e.g. 2003 YH136 q=0.25, 2010 VP21 q=0.33), so q > 1.3 is not the gating factor — they are in NEOfixer's orbital catalog but not in its observation-target list.

Group B2 — 61 genuinely unknown to NEOfixer's API

Both `/orbit/?object=` (returns "Unknown object specified", JSON-RPC code -32602) and `/obs/?object=` (returns "The specified format is not available for that object" or "Unknown object", code -3) reject these. The public web URL `https://neofixer.arizona.edu/site/500/{packed}` returns HTTP 404 for sample probes. These objects are simply not in NEOfixer's catalog as currently exposed.

Most are older NEOs (often single-apparition or short-arc); a few have very deep q and would normally be high-interest targets if NEOfixer had data on them.

packed	designation	MPC q	MPC e	MPC i (°)	obs (mpc_orbits)
J97N06J	1997 NJ6	1.1518	0.2323	18.70	265

packed	designation	MPC q	MPC e	MPC i (°)	obs (mpc_orbits)
J98H03M	1998 HM3	1.1691	0.0622	39.33	0
K01M03R	2001 MR3	1.2982	0.4522	4.42	184
K01U11Q	2001 UQ11	1.2968	0.5128	3.86	363
K04F17M	2004 FM17	0.6640	0.2500	6.76	227
K04X14P	2004 XP14	0.8851	0.1586	32.95	1399
K07X16H	2007 XH16	0.9082	0.2349	27.43	467
K08T04C	2008 TC4	0.3476	0.5545	10.65	182
K13R36F	2013 RF36	0.6506	0.7176	28.27	0
K13X28H	2013 XH28	0.5493	0.3069	19.47	14
K15Ff5E	2015 FE415	0.3338	0.7747	4.35	13
K16CW3G	2016 CG323	0.9465	0.1376	13.82	12
K16CW3H	2016 CH323	0.7863	0.2398	17.56	12
K16CW3J	2016 CJ323	0.5108	0.3197	8.70	14
K16CW3L	2016 CL323	0.7461	0.2188	5.99	20
K16CW3M	2016 CM323	0.8995	0.4242	12.73	15
K16K10G	2016 KG10	0.9995	0.5072	9.74	14
K16M04W	2016 MW4	0.3883	0.4495	12.15	24
K16N90U	2016 NU90	0.9494	0.6221	11.01	15
K16N90W	2016 NW90	1.0135	0.5266	10.65	52
K16N90Y	2016 NY90	0.4730	0.4415	0.95	25
K16UF0E	2016 UE150	0.7541	0.5742	2.08	15
K16UE9T	2016 UT149	0.7752	0.2093	2.73	21
K16UE9V	2016 UV149	0.9387	0.2668	15.40	16
K16UE9W	2016 UW149	0.8886	0.2453	10.16	24
K16UE9X	2016 UX149	0.8883	0.1938	12.65	20
K16UE9Y	2016 UY149	0.9662	0.2937	26.30	13
K16UE9Z	2016 UZ149	0.8427	0.3435	25.89	29
K16V21T	2016 VT21	0.7366	0.5672	7.44	20
K16V21V	2016 VV21	1.0159	0.6513	5.58	29
K16W58A	2016 WA58	0.7089	0.2342	12.94	15
K16W58D	2016 WD58	0.8240	0.1088	49.40	23
K16W58E	2016 WE58	0.9026	0.4349	1.14	29
K16W57O	2016 WO57	0.9987	0.5931	2.50	49
K16W57P	2016 WP57	0.3068	0.6865	0.70	20
K16W57R	2016 WR57	0.9201	0.4220	28.62	13
K16W57S	2016 WS57	0.9349	0.6243	5.30	23
K16W57T	2016 WT57	0.9210	0.6498	17.30	24
K16W57U	2016 WU57	0.3577	0.8344	3.72	17
K16W57X	2016 WX57	0.7351	0.1527	12.31	37
K16X24Z	2016 XZ24	0.7550	0.3483	40.62	18
K16Y14M	2016 YM14	0.9849	0.3198	22.51	29
K17BE2C	2017 BC142	0.9067	0.3322	15.96	31
K17C36B	2017 CB36	0.9016	0.0825	8.90	10
K17C36C	2017 CC36	0.5545	0.3038	2.10	12
K17C36D	2017 CD36	0.8171	0.2373	24.81	15
K17C36E	2017 CE36	0.8012	0.4347	5.87	36
K17C36G	2017 CG36	0.2258	0.8864	19.43	18
K17C35W	2017 CW35	0.9334	0.6197	8.72	31
K17C35X	2017 CX35	0.7047	0.4941	36.91	19
K17C35Y	2017 CY35	0.4863	0.3463	2.89	22
K17C35Z	2017 CZ35	0.7934	0.2552	14.20	15
K17DC3G	2017 DG123	0.8865	0.5514	5.00	36
K17DC3J	2017 DJ123	0.7863	0.4222	8.15	54
K17DC3K	2017 DK123	0.8690	0.2591	23.55	23
K17DC3L	2017 DL123	0.7037	0.2031	19.80	20
K17DC3O	2017 DO123	0.7743	0.4746	7.44	23
K17E25N	2017 EN25	0.4658	0.5252	29.81	23

packed	designation	MPC q	MPC e	MPC i (°)	obs (mpc_orbits)
K17E25P	2017 EP25	0.9776	0.4666	4.08	24
K20X05L	2020 XL5	0.6132	0.3872	13.85	61
K26G00B	2026 GB	0.9562	0.6177	10.23	188

Group C — 38 only-in-NEOfixer (boundary-q disagreement)

All 38 objects flagged `in_neofixer=True` but `in_mpc=False`, `in_mpc_orbits=False`, `in_cneos=False`, `in_neocc=False`, `in_lowell=False` are present in `mpc_orbits` (the table itself, joined on the unpacked primary provisional designation). They are absent from the consensus's `in_mpc_orbits` flag because **all 38 have `mpc_orbits.q > 1.3`** (range: 1.3000 to 1.3916; median ≈ 1.302). NEOfixer's Find_Orb fits give them q ≤ 1.3 (33 of the 38 currently in cache; the other 5 carry `in_neofixer=True` from today's ingest but their packed designation in the cache differs from the consensus canonical form — see *Note on canonicalisation* below).

This is the symmetric counterpart of Group A: when NEOfixer's Find_Orb and MPC's orbit solution disagree near the q = 1.3 cutoff, one side calls "NEO" and the other does not. The other four sources (NEA.txt, CNEOS, NEOCC, Lowell) all use a q ≤ 1.3 NEO definition with their own orbit catalogs, and they side with MPC here.

packed	designation	NF q	MPC q	Δ (MPC−NF)	MPC e	NF H
K12T79A	2012 TA79	—	1.3015	—	0.4405	—
K11F87R	2011 FR87	—	1.3009	—	0.4114	—
K19C00P	2019 CP	—	1.3007	—	0.2679	—
K03A23E	2003 AE23	—	1.3004	—	0.3270	—
K02S41S	2002 SS41	—	1.3003	—	0.3822	—
K13T04U	2013 TU4	1.0900	1.3507	+0.2607	0.2094	19.98
K20F07H	2020 FH7	1.0870	1.3290	+0.2420	0.2559	21.01
K13S24T	2013 ST24	1.2340	1.3916	+0.1576	0.1889	21.16
K01D77C	2001 DC77	1.2690	1.3149	+0.0459	0.4878	19.92
K18C00S	2018 CS	1.2650	1.3104	+0.0454	0.4159	20.64
K13R74A	2013 RA74	1.2810	1.3257	+0.0447	0.2861	22.49
K21B00G	2021 BG	1.2680	1.3099	+0.0419	0.5247	19.71
K05Q87N	2005 QN87	1.2660	1.3013	+0.0353	0.3543	20.17
K09S00Z	2009 SZ	1.2860	1.3058	+0.0198	0.3731	20.30
J99H01W	1999 HW1	1.2920	1.3101	+0.0181	0.4505	20.01
K22E04Q	2022 EQ4	1.2920	1.3072	+0.0152	0.5399	21.26
K06V03B	2006 VB3	1.2860	1.3003	+0.0143	0.5430	21.16
K12U27W	2012 UW27	1.2880	1.3016	+0.0136	0.5404	21.14
K09X08J	2009 XJ8	1.2910	1.3024	+0.0114	0.4714	21.04
K22B09J	2022 BJ9	1.2930	1.3027	+0.0097	0.4304	20.79
K10K80J	2010 KJ80	1.2960	1.3047	+0.0087	0.5033	19.30
K24S05C	2024 SC5	1.3000	1.3081	+0.0081	0.4028	20.35
K14G53E	2014 GE53	1.2960	1.3038	+0.0078	0.5120	18.48
K02E01H	2002 EH1	1.2950	1.3024	+0.0074	0.4207	19.56
K16L51F	2016 LF51	1.2990	1.3035	+0.0045	0.5043	18.88
K13V12Q	2013 VQ12	1.2970	1.3015	+0.0045	0.2472	20.56
K18F05H	2018 FH5	1.2980	1.3010	+0.0030	0.4922	20.35
K17F90Q	2017 FQ90	1.2990	1.3015	+0.0025	0.3629	22.17
K20X05U	2020 XU5	1.2980	1.3002	+0.0022	0.4275	21.18
K10G23J	2010 GJ23	1.2990	1.3010	+0.0020	0.5222	19.37
K13T04D	2013 TD4	1.3000	1.3019	+0.0019	0.4024	21.26
K21N50N	2021 NN50	1.3000	1.3017	+0.0017	0.5210	19.55
K20H02S	2020 HS2	1.3000	1.3009	+0.0009	0.4221	21.37
K22A06U	2022 AU6	1.3000	1.3005	+0.0005	0.0779	21.11
K04H02A	2004 HA2	1.3000	1.3005	+0.0005	0.2700	20.09
K21P23D	2021 PD23	1.3000	1.3004	+0.0004	0.3977	21.58
K22D05D	2022 DD5	1.3000	1.3002	+0.0002	0.3167	21.66

packed	designation	NF q	MPC q	Δ (MPC−NF)	MPC e	NF H
K22C12L	2022 CL12	1.3000	1.3000	+0.0000	0.1559	23.06

Note on canonicalisation

Five of the 38 (rows with `NF q = —` above) carry `in_neofixer=True` in the consensus but have no entry in today's `app/.neofixer_targets.json` keyed on the packed designation shown. Their `in_neofixer=True` flag came in via `canonicalize()` mapping the cache's `packed` key to a slightly different canonical packed designation in the consensus table. The five are: 2002 SS41, 2003 AE23, 2011 FR87, 2012 TA79, 2019 CP. A focused designation-canonicalisation audit would be a useful follow-up; it is not the main story of this disagreement.

Recommendations

1. **Drop the client-side `q > 1.3` filter** in `lib/neo_consensus_neofixer.py` (or relax it to a broader cushion such as `q > 1.5`). Membership in NEOfixer's `site=500` list is itself NEOfixer's NEO classification signal; second-guessing it with our own q test forces NEOfixer's flag to track MPC's solution rather than NEOfixer's. Recovering those 45 reduces the largest single disagreement bucket and makes the consensus tab more honestly "what does each source say".
2. **Document Group B1 as expected behaviour**, or follow up with NEOfixer to ask whether `site=500` should include orbit-only (non-priority) entries. The current bulk endpoint returns ~42K NEOs whereas the underlying orbit catalog clearly has more. If NEOfixer can expose a "full catalog" endpoint, ingesting from that would be more comparable to the other five sources.
3. **Group B2 is real catalog divergence** — keep flagging these as `in_neofixer=False`. They are not present in NEOfixer's API by any path tested. These are candidates for a future "objects each source uniquely lacks" panel on the dashboard.
4. **Group C is unavoidable boundary-class disagreement** unless we widen the NEO definition. A separate "borderline NEOs" view with $1.3 < q \le 1.4$ is worth considering for the dashboard.

Follow-up (2026-05-06): why are B1 and B2 missing from NEOfixer?

Two days after the initial analysis, this section pulls the per-object orbit quality data NEOfixer publishes on its `/orbit/` endpoint for the 15 B1 objects, joins our DB stats (arc, last observation, observation count) for the 61 B2 objects, and compares both against a 500-row random sample of objects NEOfixer *does* include — to identify the actual gating filter and to separate "NEOfixer correctly omitted" cases from "NEOfixer is missing real data".

B1 deep-dive: 15 in `/orbit/` but not in `site=500`

NEOfixer's `site=500` is operationally a *target priority list* for tonight's observation, not a complete catalog. Three filters appear to gate the cut from the underlying `/orbit/` catalog into the `/targets/?site=500` response:

1. **Orbit uncertainty** — Find_Orb's `U` parameter (0=tight, 9=lost). Median U for B1 is 7.
2. **Recency of last observation** — median 593 days since last obs for B1, with five above 1500 days (effectively lost objects).
3. **NEOfixer's own q solution** — three of the 15 have NF Find_Orb q close to or above 1.3 (1.273, 1.279, 1.730), so even without the advisory `q−max=1.3` URL parameter, NEOfixer's server-side classification deprioritises them.

Per-object table, sorted by U (high U = high uncertainty):

packed	designation	NF q	NF H	U	rms (″)	wrms	n_used	arc (d)	last obs	days ago
K12F10D	2012 FD10	1.279	21.19	9.8	0.39	0.71	31	1	2012-03-20	5160
K03YD6H	2003 YH136	0.253	19.02	9.8	0.27	0.72	46	6	2023-11-18	900
K25QE7N	2025 QN147	1.730	21.81	9.4	0.13	0.86	9	11	2025-08-28	251
K25O13O	2025 OO13	0.871	23.69	9.0	0.12	0.34	8	15	2025-08-17	262
K24R12P	2024 RP12	1.001	24.17	8.1	0.24	0.69	38	11	2024-09-20	593
K26B02E	2026 BE2	1.092	24.56	7.6	0.20	0.93	25	3	2026-01-21	105
K25D49K	2025 DK49	1.048	22.35	7.4	0.20	0.93	19	62	2025-04-28	373
K10V21P	2010 VP21	0.334	23.51	7.0	0.41	0.48	103	23	2010-11-28	5638
K25X06N	2025 XN6	0.595	29.04	6.5	0.19	0.82	18	2	2025-12-17	140
K09V09P	2009 VP9	1.273	18.43	5.8	0.24	0.55	11	4546	2022-04-20	1477
K10N81T	2010 NT81	0.961	21.69	5.1	0.62	0.67	14	4247	2022-02-26	1530
K05X04Y	2005 XY4	0.424	23.49	3.1	1.85	2.07	19	7385	2026-02-24	71
K15G00Y	2015 GY	0.854	21.68	2.1	0.39	0.56	256	662	2017-01-31	3382
K17Q17P	2017 QP17	0.623	19.58	0.6	0.40	0.61	106	2402	2024-03-18	779
K10V72D	2010 VD72	0.752	21.54	-0.0	0.59	0.81	99	5467	2025-10-28	190

Two anomalies do not fit any of the three heuristics:

- **2010 VD72** (U≈0, n_used=99, last obs 190 days ago) — well-determined orbit, recently observed, q=0.752 (potentially hazardous). No obvious reason for exclusion.
- **2017 QP17** (U=0.6, n_used=106, last obs 779 days ago) — also well-determined, q=0.623. Less recent than 2010 VD72 but still inside the typical follow-up window.

Possible additional gating: NEOfixer may exclude well-determined orbits that don't *need* more astrometry from `site=500` (treating the list as a "needs follow-up tonight" queue, not a "all known NEOs" catalog). Without internal NEOfixer documentation we can't confirm.

B2 deep-dive: 61 unknown to NEOfixer's API

B2 splits cleanly into two subpopulations once orbital arc and absolute magnitude are considered:

- **B2a — 50 short-arc faint detections** (arc < 1 day, H > 26): single-night CSS-style tracklets with provisional designations but no follow-up.
- **B2b — 11 well-observed numbered NEOs** (arc 148–9657 d, nobs 61–1399, all but one numbered).

B2a: short-arc faint single-night detections All 50 have:

- `arc_days < 1` (single-night detections, often a few hours of observations)
- H typically 27–33 (extremely faint — sub-100-metre objects observable only at closest approach)
- `last_obs` 3300–4500 days ago (no follow-up since the discovery night)

They cluster heavily by half-month, consistent with CSS post-processing precovery batches (running tracklet detection over old images, submitting designations en masse):

year + half-month	count	sample members
2016 W	10	2016 WO57, 2016 WP57, 2016 WR57, 2016 WS57, 2016 WT57, 2016 WU57, … (+4)
2017 C	9	2017 CW35, 2017 CX35, 2017 CY35, 2017 CZ35, 2017 CB36, 2017 CC36, … (+3)
2016 U	7	2016 UT149, 2016 UV149, 2016 UW149, 2016 UX149, 2016 UY149, 2016 UZ149, … (+1)
2016 C	5	2016 CG323, 2016 CH323, 2016 CJ323, 2016 CL323, 2016 CM323
2017 D	5	2017 DG123, 2017 DJ123, 2017 DK123, 2017 DL123, 2017 DO123
2016 N	3	2016 NU90, 2016 NW90, 2016 NY90

This is NEOfixer doing the right thing. With only single-night astrometry from a decade ago, Find_Orb cannot fit a usable orbit, and even if it could, the prediction uncertainty after 9–10 years would dwarf the celestial sphere. Listing these on a site=500 target page would direct observers to chase impossible-to-find ghosts. The other five sources (MPC NEA.txt, mpc_orbits, CNEOS, NEOCC, Lowell) keep them in their catalogues for historical record-keeping, not as observation targets.

B2b: well-observed numbered NEOs absent from NEOfixer (real catalog gap) These are ordinary, multi-decade-arc NEOs with hundreds of observations, mostly numbered. NEOfixer should have them; it does not.

packed	designation	permid	nobs	arc (d)	last obs	mpc q	mpc H
`J97N06J`	1997 NJ6	189011	265	9657	2023-12-16	1.152	19.08
`J98H03M`	1998 HM3	326291	586	9172	2023-06-01	1.169	19.00
`K01M03R`	2001 MR3	333311	184	8148	2023-10-12	1.298	19.07
`K01U11Q`	2001 UQ11	306918	363	8093	2023-11-18	1.297	17.31
`K04F17M`	2004 FM17	387816	227	7351	2024-05-08	0.664	19.36
`K07X16H`	2007 XH16	484402	465	6490	2025-09-14	0.908	19.68
`K04X14P`	2004 XP14	612901	1399	5785	2020-10-12	0.885	19.79
`K08T04C`	2008 TC4	614134	182	4726	2021-09-14	0.348	21.65
`K20X05L`	2020 XL5	614689	61	4390	2024-12-30	0.613	20.28
`K13R36F`	2013 RF36	555122	123	4250	2025-04-25	0.651	16.97
`K26G00B`	2026 GB	—	188	148	2026-04-25	0.956	22.51

I probed NEOfixer for 9 of these 11 across every identifier the API accepts — packed designation, unpacked designation, permid as a bare integer string — and the public webpage `https://neofixer.arizona.edu/site/500/{packed}`. Every probe returned "Unknown object specified." (JSON-RPC code -32602) or HTTP 404. **2008 TC4** alone has a famous 2017 close approach and 1399 catalogued observations; its absence is striking.

Plausible explanations (none confirmed without insight into NEOfixer's ingestion):

1. NEOfixer's Find_Orb may have generated a non-NEO solution for these (a different orbit fit), causing them to be classified out of NEOfixer's NEO database. Unlikely for objects with thousands of observations and tight U=0 orbits, but possible if NEOfixer started from a different astrometry slice.
2. NEOfixer's ingestion may have specifically deprioritised these (recovered, numbered, well-determined — "no new astrometry needed"). But B1 has well-determined numbered objects that *are* in `/orbit/` if not `/targets/`, so a complete API drop is more drastic than mere deprioritisation.
3. A specific catalog-version or sync issue: these may have been removed during a maintenance cycle and not re-added.

Worth a follow-up email to Eric Christensen / Carson Fuls (NEOfixer maintainers) with this list and a request for clarification.

What is NEOfixer doing right vs wrong, compared to the other five sources?

The starting frame matters: **NEOfixer is an observation-prioritisation tool, not a NEO catalogue.** The other five (MPC NEA.txt, mpc_orbits, CNEOS, NEOCC, Lowell astorb) are catalogues, listing every known NEO regardless of observability. Once that asymmetry is acknowledged, much of the "missing" gets reframed:

Behaviour	NEOfixer	Catalogues
Single-night unrecoverable detections (B2a, 50 objects)	Excludes	Include
Well-determined orbits, lost or low-priority (B1, 13 of 15)	In `/orbit/`, not `site=500`	Include
Boundary-q disagreements (Group A, 45)	Trusts own Find_Orb	Trust own orbits
Independent orbit fitting	Yes (Find_Orb)	MPC fits, others import
Famous well-observed NEOs (B2b, 11)	**Absent — gap**	Include

NEOfixer doing right:

- **Filters out 50 single-night unrecoverable tracklets**. These are valid astronomical detections with no observational utility tonight; chasing them would waste telescope time. The other catalogues retain them for record-keeping; that's appropriate for catalogues but inappropriate for a target list.
- **Maintains independent Find_Orb solutions**. The Group A and Group C disagreements in the original analysis (83 of 159 total disagreements) are NEOfixer's Find_Orb saying something different from MPC at the q≈1.3 boundary. That's genuine cross- validation; without NEOfixer the consensus tab would have less analytical value.

NEOfixer doing wrong:

- **Catalog gap for 11 well-observed numbered NEOs (B2b)**. No astronomical reason — these have multi-decade arcs and hundreds of observations. Best guess is an ingestion-pipeline gap; should be raised with NEOfixer maintainers.
- **Inconsistency between `/orbit/` and `/targets/`** for B1. The dashboard's per-MPEC link to `https://neofixer.arizona` returns HTTP 404 for these 15 objects, which is a mild user-facing surprise. NEOfixer could expose a "full catalog" endpoint or document the gating logic.

Updated recommendations

To the original four, add:

5. **Recover B1 by ingesting from `/orbit/` per-object** for designations that are in the other-five set but missing from `/targets/?site=500`. ~15 calls per refresh, throttled. This recovers NEOfixer-confirmed orbits that the bulk endpoint omits.
6. **Email NEOfixer maintainers** with the B2b list (8 numbered NEOs probed; 3 more in the table) and ask whether the absences are intentional or reflect an ingestion gap. Cite specific examples: 2008 TC4 (614134), 2004 XP14 (612901), 1998 HM3 (326291).
7. **Document NEOfixer as a prioritisation tool** in the consensus tab's legend. Currently the six sources are presented symmetrically; users may interpret `in_neofixer=False` as "NEOfixer disagrees", when frequently it just means "NEOfixer has no priority for this object tonight". A small label change would reduce that confusion.